Spiral similarity and the configuration toolkit

Kevin Hu
November 2026

L5-GE2 Level 5 Advanced · Geometry

Spiral similarity centres, the Miquel point and its inversion, symmedians, isogonal conjugates and the mixtilinear incircle, proved and used as tools.
By the end you can find the centre of a spiral similarity and the Miquel point of four lines; recognise symmedians and isogonal conjugates and prove their standard properties; use the incentre-excentre lemma, the shooting lemma and mixtilinear facts in full proofs.
Prerequisites: Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2), Inversion and projective methods (L5-GE1). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Let ABC be a triangle with $AB \neq AC$. Points D and E move on sides AB and AC so that $BD = CE$. Prove that the circle ADE passes through a fixed point other than A. A drawing suggests the midpoint N of arc BAC, but angle chasing at N does not start, because nothing yet relates N to D and E. The relation is that N is the centre of a rotation taking D to E and B to C.

Directed angles $\measuredangle$ modulo 180°, the ratio lemma, the incentre-excentre lemma (L4-GE1, L4-GE2), inversion and Brocard's theorem (L5-GE1) are assumed. Throughout, Γ is the circumcircle of triangle ABC and O is its centre.

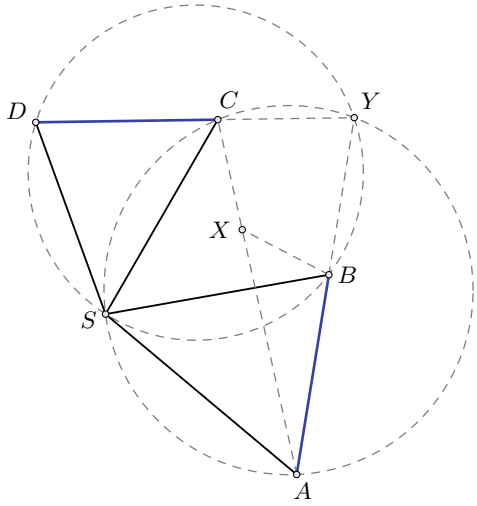
1.1 Spiral similarity

A *spiral similarity* with centre S, angle θ and ratio $k > 0$ is the rotation about S through θ followed by the homothety at S with ratio k. As a map of complex numbers this is $z \mapsto s + ke^{i\theta}(z - s)$. Such a map turns every line through the angle θ, and two spiral similarities with the same centre commute.

Theorem 1.1.
If $A \neq B$, $C \neq D$ and $\overrightarrow{AB} \neq \overrightarrow{CD}$, exactly one spiral similarity sends A to C and B to D.
Theorem 1.2. Let S be the centre of the spiral similarity $A \mapsto C$, $B \mapsto D$.
(1) If lines AB and CD meet at Y, then S, Y, A, C are concyclic, and so are S, Y, B, D.
(2) S is also the centre of the spiral similarity $A \mapsto B$, $C \mapsto D$. Hence, if lines AC and BD meet at X, then S lies on the circles XAB and XCD.

Proof. For 1.1, the map is $z \mapsto \alpha z + \beta$ with $\alpha = \frac{d-c}{b-a} \neq 1$, and its only fixed point is $s = \beta/(1-\alpha)$. For 1.2(1), the map turns line AB onto line CD through θ, so $\measuredangle(YA, YC) = \theta = \measuredangle(SA, SC)$, and the same holds for B, D. (If $Y = A$, read the first circle as the circle through A, C, S touching line AB at A.) For (2), $SC/SA = SD/SB$ and the oriented angles ASC, BSD both equal θ. Hence $SB/SA = SD/SC$ and the oriented angles ASB, CSD are equal, so the spiral similarity at S taking A to B takes C to D. Now apply (1) to this map. ■

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The opening problem. Lines DB and EC are not parallel, so there is a spiral similarity $D \mapsto E$, $B \mapsto C$. These lines meet at A, so by 1.2(1) the centre S lies on the circles ADE and ABC. The ratio is $EC/DB = 1$, so the map is a rotation and $SB = SC$. The rotation takes ray BA to ray CA, so the oriented angle BSC equals the oriented angle BAC, and S lies on the same arc BC as A. Hence $S = N$.

Corollary 1.3.
Circles ω_1, ω_2 with centres O_1, O_2 meet at P and Q. A line through P meets ω_1 again at A and ω_2 again at B. Then the spiral similarity with centre Q sending A to B is the same for every such line. This map sends ω_1 to ω_2, O_1 to O_2, and the tangent at A to the tangent at B.
Lemma 1.4 (fixed shape). Let S be a fixed point, and let P' depend on P so that triangle SPP' is directly similar to a fixed triangle. Then $P \mapsto P'$ is a spiral similarity with centre S. So if P moves on a line or a circle, then so does P'.

Proof. Lemma 1.4 holds because $p' - s = \lambda(p - s)$ for a fixed complex number λ. For 1.3, as the line turns about P, the directed angles $\measuredangle QAB = \measuredangle QAP$ and $\measuredangle ABQ = \measuredangle PBQ$ stay constant, so all triangles QAB are directly similar. By Lemma 1.4, $A \mapsto B$ is one spiral similarity with centre Q, and this map sends ω_1 onto ω_2. A similarity sends centres to centres and tangents to tangents. ■

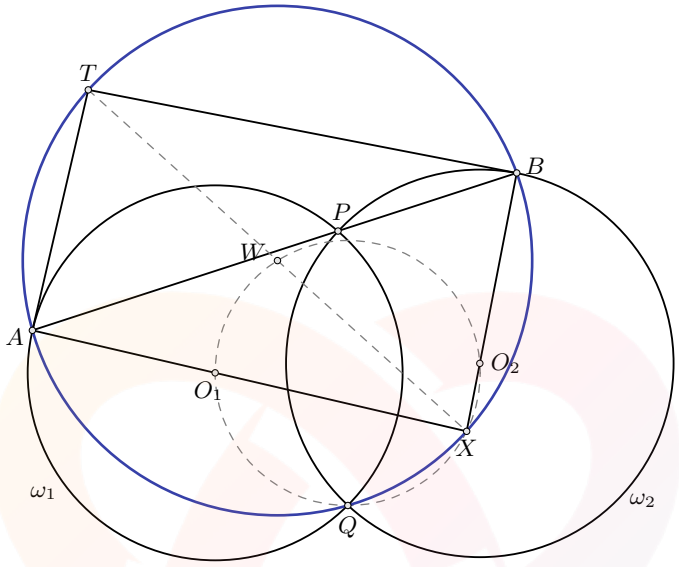
Example 1.
Circles ω_1 and ω_2, with centres O_1 and O_2, meet at P and Q. A line through P meets ω_1 again at A and ω_2 again at B. The tangent to ω_1 at A and the tangent to ω_2 at B meet at T, and lines AO_1 and BO_2 meet at X. Prove that TX is a diameter of the circle ABQ, and that the midpoint of TX lies on the circle QO_1O_2.

Solution.
Let σ be the spiral similarity of Corollary 1.3. Its angle is $\measuredangle AQB \neq 0$, since Q is not on line AB. The map σ sends the tangent at A to the tangent at B, and line AO_1 to line BO_2. So T and X exist, and by Theorem 1.2(1) both lie on the circle ABQ.
Line AB meets ω_2 twice, so A is not on the tangent at B and $T \neq A$. The point X cannot equal both A and B, say $X \neq A$. Then $\angle TAX = 90^\circ$, because a tangent is perpendicular to the radius. Hence TX is a diameter of the circle ABQ, and the midpoint W of TX is the circumcentre of triangle QAB.
By Theorem 1.2(2), Q is the centre of a spiral similarity τ with $A \mapsto O_1$, $B \mapsto O_2$. The map

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τ sends triangle QAB to triangle QO_1O_2, so τ sends W to the circumcentre Z of triangle QO_1O_2. By 1.2(2) again, triangles QAO_1 and QWZ are directly similar. As $O_1A = O_1Q$, we get $ZW = ZQ$, so W lies on the circle QO_1O_2.

Remark. The points T and X are cut out by lines through A and B that correspond under σ, so both lie on the circle ABQ before any angle is chased. Triangle QAB has the fixed shape of QO_1O_2, so we compare the two circumcentres.



1.2 The Miquel point

Theorem 2.1 (Miquel).
Let $ABCD$ be a quadrilateral with no two sides parallel, $Q = AB \cap CD$ and $R = AD \cap BC$. The circles QAD, QBC, RAB, RCD pass through one point M, the Miquel point. This point is the centre of the spiral similarities $A \mapsto B$, $D \mapsto C$ and $A \mapsto D$, $B \mapsto C$.

Theorem 2.2. If moreover $ABCD$ is convex and cyclic with centre O, then M lies on QR, $OM \perp QR$, and $QR^2 = QA \cdot QB + RA \cdot RD$.

Theorem 2.3. In Theorem 2.1, $MA \cdot MC = MB \cdot MD = MQ \cdot MR$, and the angles AMC, BMD, QMR have the same internal bisector ℓ. Hence the reflection in ℓ followed by the inversion with centre M and radius $\sqrt{MA \cdot MC}$ swaps A with C, B with D and Q with R.

Proof of 2.1. Let M be the centre of the spiral similarity $A \mapsto B$, $D \mapsto C$, which exists because $AD \nparallel BC$. Lines AD and BC meet at R, so M lies on the circles RAB and RDC by 1.2(1). Lines AB and DC meet at Q, so M lies on the circles QAD and QBC by 1.2(2), which also gives the second spiral similarity. ■

Proof of 2.2. From the circles $QBCM$ and $RDCM$,

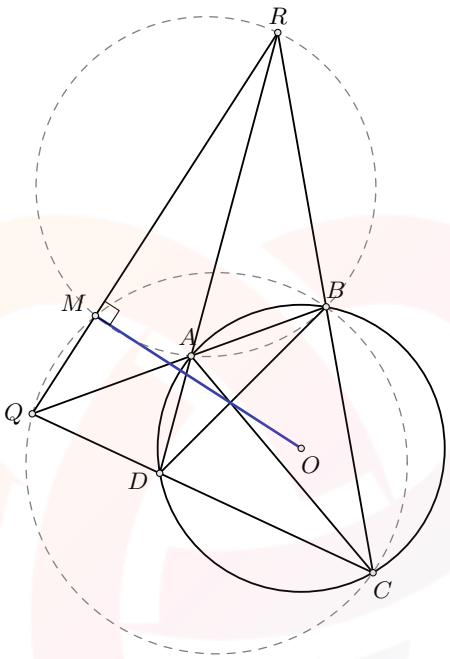
$$\measuredangle QMR = \measuredangle QMC + \measuredangle CMR = \measuredangle QBC + \measuredangle CDR = \measuredangle ABC + \measuredangle CDA = 0,$$

so Q, M, R are collinear. The powers of Q in circle $RABM$ and of R in circle $QADM$ give $QA \cdot QB = QM \cdot QR$ and $RA \cdot RD = RM \cdot RQ$, signed along line QR. Adding gives the formula for QR^2. Subtracting, and writing the two powers as $OQ^2 - \rho^2$ and $OR^2 - \rho^2$, gives $OQ^2 - OR^2 = MQ^2 - MR^2$, so $OM \perp QR$. ■

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Proof of 2.3. Triangles MAB and MDC are directly similar by 2.1. So $MA/MD = MB/MC$, and the oriented angle from ray MA to ray MB equals the oriented angle from ray MD to ray MC. If the four rays have arguments a, b, c, d, this says $a + c \equiv b + d$ modulo $360°$, and the internal bisector of angle AMC has direction $\frac{1}{2}(a + c)$ modulo $180°$. Hence $MA \cdot MC = MB \cdot MD$ and the two bisectors agree. The quadrilateral $AQCR$ has the same four side lines, so the same Miquel point, and its pairs of opposite vertices are A, C and Q, R. The same argument gives the statement for Q, R. ■

In the cyclic case ℓ is line MO, and M is the inverse of $P = AC \cap BD$ in the circumcircle (L5-GE1, Theorem 4.4).



Example 2.
Let $ABCD$ be a convex quadrilateral with no two sides parallel, let $Q = AB \cap CD$ and $R = AD \cap BC$, and let M be the Miquel point. Suppose that M lies on neither of the lines AC and QR. Prove that the circles MAC and MQR touch each other if and only if the diagonal BD bisects the diagonal AC.

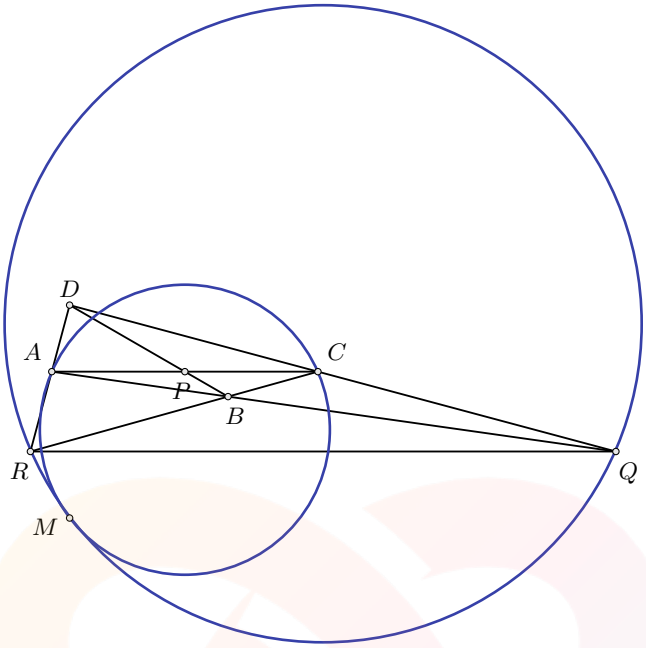
Solution.
Let Ψ be the map of Theorem 2.3. Line AC does not pass through M, so Ψ sends line AC to a circle through M, and this circle contains $\Psi(A) = C$ and $\Psi(C) = A$. Hence Ψ swaps line AC with the circle MAC, and in the same way line QR with the circle MQR.
Two different circles through M touch if and only if M is their only common point. As Ψ is a bijection of the plane without M, this happens if and only if lines AC and QR have no common point. So the circles touch if and only if $AC \parallel QR$.
Let $P = AC \cap BD$. In triangle BAC the points Q and R lie on lines BA and BC, and the cevians AR, CQ, BP pass through D. Ceva's theorem with signed ratios gives

$$\frac{AP}{PC} \cdot \frac{CR}{RB} \cdot \frac{BQ}{QA} = 1.$$

So $AP = PC$ if and only if $\frac{BQ}{QA} = \frac{BR}{RC}$, that is, if and only if $QR \parallel AC$.
Remark. Circles touching at the centre of an inversion become parallel lines (L5-GE1, Sec-

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tion 2). Here Ψ turns the two circles into two diagonals of the complete quadrilateral.



1.3 Symmedians

Two lines through A that are mirror images in the bisector of angle A are called *isogonal*. The *A-symmedian* is the line isogonal to the median AM.

Theorem 3.1.
Let the A-symmedian meet BC at L and Γ again at K.
(1) $BL/LC = AB^2/AC^2$.
(2) $KB/KC = AB/AC$, and K is the only point $X \neq A$ of Γ with $XB/XC = AB/AC$.
(3) If $\angle A \neq 90^\circ$, the tangents to Γ at B and C meet on line AK.
(4) The symmedian bisects every segment XY with X on line AB, Y on line AC and B, C, Y, X concyclic (an antiparallel to BC).
Theorem 3.2. The midpoint S of the chord AK is the centre of the spiral similarity $B \mapsto A$, $A \mapsto C$. So triangles SBA and SAC are directly similar, $SA^2 = SB \cdot SC$, the circle SAB touches AC at A, and the circle SAC touches AB at A.

Proof of 3.1. The ratio lemma gives $\frac{BL}{LC} = \frac{AB \sin \angle BAL}{AC \sin \angle LAC}$ and, for the median, $1 = \frac{AB \sin \angle BAM}{AC \sin \angle MAC}$. Isogonality swaps the two sines, which proves (1). For (2), $\sin \angle BAK = KB/2R$ and $\sin \angle KAC = KC/2R$, so (1) gives $KB/KC = AB/AC$. The points X with $XB/XC = AB/AC$ form a circle or a line, which meets Γ at most twice. For (3), let the tangents meet at T and let line AT meet Γ again at K'. From $TB^2 = TK' \cdot TA$, triangles TBK' and TAB are similar, so $BK'/AB = TB/TA$. Likewise $CK'/AC = TC/TA$. As $TB = TC$, we get $K'B/K'C = AB/AC$ and $K' = K$ by (2). For (4), reflection in the bisector of angle A turns antiparallels into parallels to BC and the symmedian into the median. ■

Proof of 3.2. Let S' be the centre of the spiral similarity $B \mapsto A$, $A \mapsto C$, with angle θ, and let K' be the reflection of A in S'. Then $S'B/S'A = S'A/S'C = AB/AC$. Ray $S'K'$ is opposite to ray $S'A$,

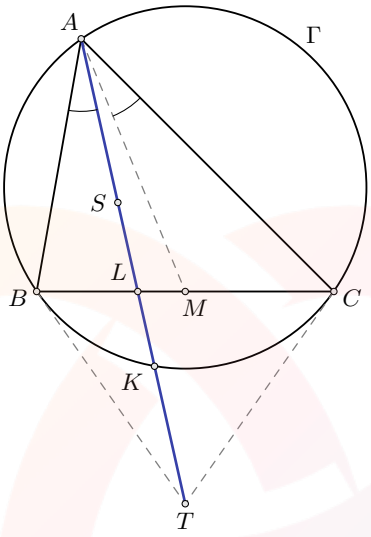
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so the rotations from ray $S'B$ to ray $S'K'$ and from ray $S'K'$ to ray $S'C$ both equal $\theta + 180°$. With $S'K' = S'A$ this makes triangles $S'BK'$ and $S'K'C$ directly similar. Hence

$$\measuredangle BK'C = \measuredangle BK'S' + \measuredangle S'K'C = \measuredangle BK'S' + \measuredangle S'BK' = \measuredangle BS'K' = \theta = \measuredangle BAC,$$

so K' lies on Γ. Also $K'B/K'C = S'B/S'K' = AB/AC$, so $K' = K$ by 3.1(2) and $S' = S$. The tangencies are $\measuredangle SBA = \measuredangle SAC$ and $\measuredangle SCA = \measuredangle SAB$. ■

Statement 3.1(3) is the form in which symmedians usually appear. You are given tangents at two vertices, and you should recognise the symmedian from the third.



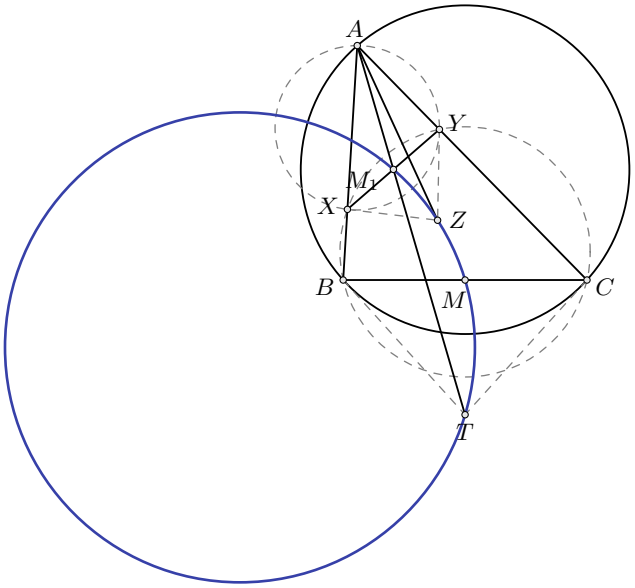
Example 3.
Let ABC be a triangle with $AB \neq AC$ and $\angle A \neq 90°$. A circle through B and C meets the sides AB and AC again at X and Y. The tangents to Γ at B and C meet at T, and the tangents to the circle AXY at X and Y meet at Z. Prove that T, Z and the midpoints of BC and XY lie on a circle.

Solution.
Let M and M_1 be the midpoints of BC and XY. Since B, C, Y, X are concyclic, $AX \cdot AB = AY \cdot AC$. Put $k = AX/AC = AY/AB$, and let ρ be the reflection in the bisector of angle A followed by the homothety at A with ratio k. Then ρ sends C to X and B to Y, so ρ sends Γ to the circle AXY, M to M_1, and T to Z.
By Theorem 3.1(3), T lies on the A-symmedian. Hence $Z = \rho(T)$ lies on the median AM, and $M_1 = \rho(M)$ lies on the symmedian AT, with $AM_1 = k \cdot AM$ and $AZ = k \cdot AT$. Therefore

$$AM_1 \cdot AT = k \cdot AM \cdot AT = AM \cdot AZ.$$

The point T lies on ray AM_1 when angle A is acute and on the opposite ray when angle A is obtuse, and the same holds for Z and ray AM, because ρ maps line AT to line AZ. So the signed products are equal. Lines AT and AM are different, as $AB \neq AC$. By the converse of the power of a point, M_1, T, M, Z are concyclic.
Remark. The segment XY is antiparallel to BC, so triangle AXY with everything attached is a reflected and scaled copy of triangle ACB. Under this copy medians and symmedians swap.

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1.4 Isogonal conjugates

Theorem 4.1.
Let P be a point not on Γ and not a vertex, and let P_a, P_b, P_c be the reflections of P in BC, CA, AB. The lines isogonal to AP, BP, CP at the three vertices pass through the circumcentre P^* of triangle $P_aP_bP_c$, the *isogonal conjugate* of P. The feet of the perpendiculars from P and from P^* to the three side lines lie on one circle, whose centre is the midpoint of PP^*.

Proof. The points P_a, P_b, P_c are collinear only when P lies on Γ (Simson line), so P^* exists. Reflection in AC followed by reflection in AB is the rotation about A through twice the angle from AC to AB, and this rotation takes P_b to P_c. So $AP_b = AP_c$, and the perpendicular bisector m of P_bP_c passes through A with $\measuredangle(AP_b, m) = \measuredangle(AC, AB)$. Since $\measuredangle(AC, AP_b) = \measuredangle(AP, AC)$, we get

$$\measuredangle(AC, m) = \measuredangle(AP, AC) + \measuredangle(AC, AB) = \measuredangle(AP, AB).$$

So m is the line isogonal to AP. The same holds at B and C, and the three perpendicular bisectors meet at P^*.

The homothety at P with ratio $\frac{1}{2}$ sends P_a, P_b, P_c to the feet D, E, F from P, and P^* to the midpoint N of PP^*. So D, E, F lie on the circle with centre N and radius $\frac{1}{2}P^*P_a$. Isogonality is symmetric, so P is the isogonal conjugate of P^*, and the feet from P^* lie on the circle with centre N and radius $\frac{1}{2}PP_a^*$, where P_a^* is the reflection of P^* in BC. Reflection in BC gives $P^*P_a = PP_a^*$. ■

Since $EF \parallel P_bP_c$, the proof also shows $AP^* \perp EF$. The pairs to know are $I \leftrightarrow I$, $O \leftrightarrow H$ (the common circle is the nine-point circle), and the centroid with the point where the three symmedians meet. Two vertices suffice to recognise a pair. If $\angle PBA = \angle QBC$ and $\angle PCA = \angle QCB$ for P, Q inside the triangle, then $Q = P^*$.

1.5 The incentre and the mixtilinear incircle

Write M_A, M_B, M_C for the midpoints of the arcs BC, CA, AB of Γ not containing the opposite vertex, and N for the midpoint of arc BAC. Recall the incentre-excentre lemma (L4-GE1), $M_AB =$

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$M_AC = M_AI = M_AI_A$. The *A-mixtilinear incircle* ω_A is the circle tangent to segments AB, AC and internally tangent to Γ. Let ω_A touch AB, AC, Γ at K, L, T.

Theorem 5.1 (shooting lemma).
If a circle inside Γ touches Γ at T and a chord XY at Z, then line TZ passes through the midpoint W of arc XY not containing T, and $WZ \cdot WT = WX^2$.

Theorem 5.2. (1) I is the midpoint of KL, and $KL \perp AI$.
(2) B, K, I, T are concyclic, and so are C, L, I, T.
(3) Line TI passes through N.
(4) If the A-excircle touches BC at X, then AT and AX are isogonal.

Proof of 5.1. The homothety at T taking the small circle to Γ takes line XY to the parallel tangent of Γ on the far side from T, and Z to the point of contact, which is W. Then $\angle WXZ = \angle WTX$ (equal arcs WX, WY), so triangles WXZ and WTX are similar and $WX^2 = WZ \cdot WT$. ■

Proof of 5.2. (1) By 5.1 with chord AB, the points T, K, M_C are collinear and $M_CK \cdot M_CT = M_CA^2 = M_CI^2$. So triangles M_CIK and M_CTI are similar with opposite orientation, and $\measuredangle(IK, IM_C) = \measuredangle(TM_C, TI)$. In the same way $\measuredangle(IM_B, IL) = \measuredangle(TI, TM_B)$. Adding,

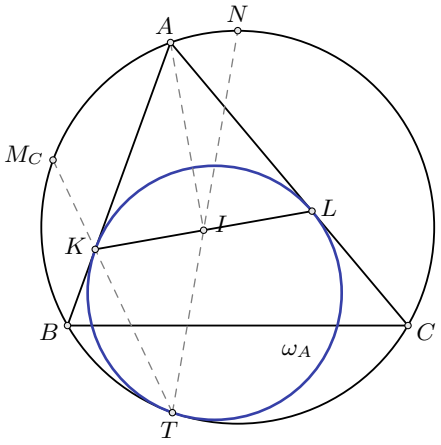
$$\measuredangle(IK, IL) = \measuredangle M_CTM_B + \measuredangle M_CIM_B = \measuredangle M_CAM_B + \measuredangle M_CIM_B.$$

Since $M_BA = M_BI$ and $M_CA = M_CI$, line M_BM_C is the perpendicular bisector of AI, so the right side is 0 and K, I, L are collinear. As $AK = AL$, statement (1) follows.

(2) Triangle AIK has a right angle at I, so $AK = AI/\cos\frac{A}{2} = bc/s$, which is less than c. Thus K lies on side AB, and $\angle BKI = 90^\circ + \frac{A}{2}$, $\angle KBI = \frac{B}{2}$ give $\angle KIB = \frac{C}{2}$. Also $\angle KTB = \angle M_CTB = \angle M_CCB = \frac{C}{2}$. As I and T lie on the same side of AB, the points B, K, I, T are concyclic.

(3) By (2), $\measuredangle BTI = \measuredangle BKI = \measuredangle(AB, KL)$ and $\measuredangle ITC = \measuredangle ILC = \measuredangle(KL, AC)$. These are equal, because reflection in AI swaps AB with AC and fixes line KL. So TI bisects angle BTC. The incentre lies inside this angle, so TI is the internal bisector, which passes through the midpoint N of the arc BC not containing T.

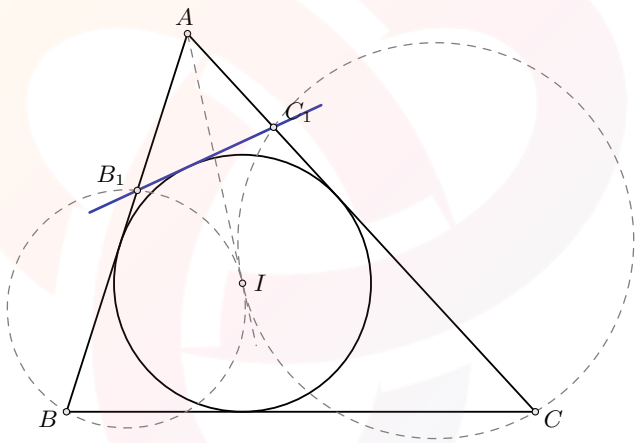
(4) Let Φ be the $\sqrt{bc}$ inversion of L5-GE1, Section 2. The map Φ swaps Γ with line BC and line AB with line AC, and sends points inside Γ to the side of BC away from A. So $\Phi(\omega_A)$ touches AB, AC and BC beyond BC. This is the A-excircle, and $\Phi(T) = X$. ■



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Example 4.
Let ABC be a triangle with incentre I. The circle through B tangent to line AI at I meets AB again at B_1, and the circle through C tangent to line AI at I meets AC again at C_1. Prove that line B_1C_1 is tangent to the incircle.

Solution.
The power of A with respect to the two circles gives $AB \cdot AB_1 = AI^2 = AC \cdot AC_1$. Invert about A with radius AI. Then B and B_1 swap, C and C_1 swap, and line B_1C_1 is mapped to the circle through A, B, C, which is Γ.
Let the incircle touch AB, AC at F, E, and let the perpendicular to AI at I meet AB, AC at K, L. In the right-angled triangle AIK the altitude from I is IF, so $AF \cdot AK = AI^2$. Hence the inversion swaps F with K and E with L, and maps the incircle to the circle touching AB at K and AC at L (L5-GE1, Theorem 1.3). By Theorem 5.2(1) this circle is ω_A.
The circle ω_A touches Γ. Inversion preserves tangency, so the incircle touches line B_1C_1.
Remark. The two tangent circles say that B_1C_1 is the inverse of Γ in the circle with centre A and radius AI. So we ask what this inversion does to the incircle.



Common traps.

- Pairing points wrongly in Theorem 1.2. The circle passes through a point, its image, and the meeting point of a line with the image line.
- Using Theorem 3.1(3) when the angle is 90° and the tangents are parallel.
- Using the shooting lemma with the arc midpoint on the same side of the chord as T.
- Quoting mixtilinear facts without proof. Beyond the incentre-excentre lemma, prove what you use.

1.6 How to spot this in a contest

1. Two circles and a line through one common point. The other common point is a spiral centre.
2. Equal ratios or equal lengths on two non-parallel segments. Find the spiral similarity, then use Lemma 1.4.
3. Four lines. Find the Miquel point first, then try the map of Theorem 2.3.

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4. Tangents at two vertices of an inscribed triangle give a symmedian. Equal angles $\angle BAX = \angle YAC$ mean isogonal lines.
5. An incentre with an arc midpoint, or a circle tangent to two sides and to Γ. Use Theorems 5.1 and 5.2.

These configurations appear throughout *Beyond the Incenter*.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆⋆ Let ABC be a triangle with $\angle A \neq 90^\circ$. The perpendicular bisector of AB meets line AC at Q, and the perpendicular bisector of AC meets line AB at P. Points D and E, other than the vertices, lie on the sides AB and AC so that

$$\frac{AD}{DB} = \frac{CE}{EA}.$$

The tangents to the circle ADE at D and E meet at Z. Prove that Z lies on line PQ.

Problem 2 ⋆⋆⋆ Let $ABCD$ be a convex quadrilateral with no two sides parallel and $AB \cdot CD = AD \cdot BC$. Lines AB and CD meet at Q, lines AD and BC meet at R, and the circles QAD and RAB meet again at M. Suppose that M lies on neither of the lines AC and BD. The A-symmedian of triangle ABD meets BD at L. Prove that line ML bisects both of the angles BMD and AMC.

Problem 3 ⋆⋆⋆ Let ABC be a triangle with $AB \neq AC$, $\angle A \neq 90^\circ$ and circumcircle Γ. A circle with centre O' passing through B and C meets the sides AB and AC again at X and Y. The circle AXY meets Γ again at W. The tangents to Γ at B and C meet at T, and the tangents to the circle AXY at X and Y meet at Z. Prove that O', T, Z, W lie on a circle.

Problem 4 ⋆⋆⋆⋆ Let ABC be a triangle with $AB \neq AC$ and circumcircle Γ, and let the A-symmedian meet Γ again at K. A circle with centre O' passing through B and C meets the sides AB and AC again at X and Y. Lines XY and BC meet at R. Prove that $\angle O'KR = 90^\circ$.

Problem 5 ⋆⋆⋆⋆ Let ABC be a triangle with incentre I, and let the incircle touch BC at D. The A-mixtilinear incircle touches AB, AC and the circumcircle at K, L and T. Points V and U lie on the side BC with $BV = BK$ and $CU = CL$. Prove that the lines IU and IV are tangent to the circle TUV, and that $\angle VTD = \angle ITU$.

Problem 6 ⋆⋆⋆⋆ Let ABC be an acute triangle. Points P and Q inside the triangle satisfy $\angle PBA = \angle QBC$, $\angle PCA = \angle QCB$ and $AP \neq AQ$. Let E and F be the feet of the perpendiculars from P to CA and AB, and let E' and F' be the feet of the perpendiculars from Q to CA and AB. Lines EF' and $E'F$ meet at X. Prove that X lies on line PQ.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

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1. **BMO2 2022/23 Problem 1** ⋆⋆
   The centres of circles ABI and ACI are arc midpoints of the circumcircle.
2. **IMO 2006 Problem 1** ⋆⋆
   Show that P lies on circle BIC and use the centre of that circle.
3. **BMO2 2012/13 Problem 2** ⋆⋆⋆
   The conclusion says AP and AQ are isogonal. Transfer the equal angles through the parallelogram.
4. **BMO2 2014/15 Problem 3** ⋆⋆⋆
   Shooting lemma at the point of tangency, then the incentre-excentre lemma in triangle AQP.
5. **BMO2 2020/21 Problem 3** ⋆⋆⋆
   K is the centre of a spiral similarity taking the incircle contact points on AB and AC to B and C.
6. **BMO2 2023/24 Problem 3** ⋆⋆⋆
   The line through the two midpoints is the radical axis of the circumcircle and the point circle at P.
7. **BMO2 2024/25 Problem 2** ⋆⋆⋆
   Both circles pass through I, and so do lines BP and CQ. Corollary 1.3 applies.
8. **IMO 2005 Problem 5** ⋆⋆⋆
   Equal lengths BC = DA and BE = DF suggest a rotation. Find the segments that the rotation maps onto each other.
9. **USAMO 2013 Problem 1** ⋆⋆⋆
   The three circles meet at the Miquel point. Look at the spiral similarities centred there.
10. **USAMO 2006 Problem 6** ⋆⋆⋆
    Consider the Miquel point of lines AB, BC, CD, DA and what the ratio condition does to this point.
11. **USAMO 2019 Problem 2** ⋆⋆⋆⋆
    Locate P through the products AP times AB and BP times AB. A symmedian of triangle AEB appears.
12. **IMO 2004 Problem 5** ⋆⋆⋆⋆
    Read both angle conditions in triangle BCD, at the vertices B and D. Section 4 is relevant.

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4 Hints

1. Some spiral similarity takes D to E for every position of D. Find its centre S, show that the circle ADE passes through S, and note that triangle SDE has a fixed shape.
2. Identify M. Two spiral similarities centred at M give MB/MA and MD/MA as ratios of sides. Compare MB/MD with BL/LD.
3. Which spiral similarity has centre W? Find the images of the circle AXY, of Z and of the perpendicular bisector of XY.
4. Two points of the circle on diameter $O'R$ are easy to name: the midpoint of XY and the Miquel point of $BCYX$. Compare powers of A.
5. Reflect K in BI and L in CI. Show that U lies on the circle $BKIT$ and V on the circle $CLIT$, then compute angle UTV.
6. The four feet lie on a circle whose centre is on PQ. Find the polar of X in this circle, using the circles on diameters AP and AQ.

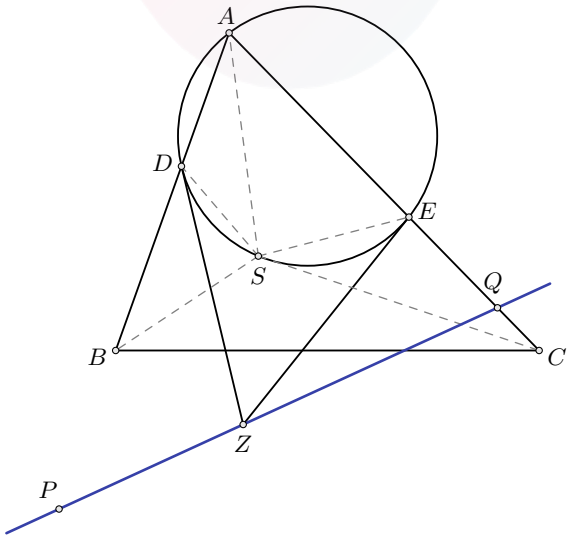
5 Solutions

Problem 1

Let ABC be a triangle with $\angle A \neq 90°$. The perpendicular bisector of AB meets line AC at Q, and the perpendicular bisector of AC meets line AB at P. Points D and E, other than the vertices, lie on the sides AB and AC so that

$$\frac{AD}{DB} = \frac{CE}{EA}.$$

The tangents to the circle ADE at D and E meet at Z. Prove that Z lies on line PQ.



Let σ be the spiral similarity $B \mapsto A$, $A \mapsto C$, with centre S. The map σ sends segment BA onto segment AC and preserves ratios, so $BD/DA = AE/EC$ gives $\sigma(D) = E$. The point S is not on line AB. Otherwise S would also lie on $\sigma(AB) = AC$, so $S = A$, but $\sigma(A) = C$.

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Step 1. The circle ADE passes through S. The map σ takes B, D on line AB to A, E on line AC, and these lines meet at A. By Theorem 1.2(1), S, A, D, E are concyclic.

Step 2. Two tangent circles. Triangles SBA and SAC are directly similar, so $\measuredangle SBA = \measuredangle SAC$ and $\measuredangle SCA = \measuredangle SAB$. By the converse of the tangent-chord theorem, the circle SBA touches AC at A and the circle SAC touches AB at A. As $\angle A \neq 90^\circ$, the chord AB is not a diameter of the circle SBA, so the tangent at B meets the tangent AC at a point equidistant from A and B. This point is Q. In the same way the tangents to the circle SAC at A and C meet at P. Since σ sends the circle SBA to the circle SAC and B, A to A, C, we get $\sigma(Q) = P$.

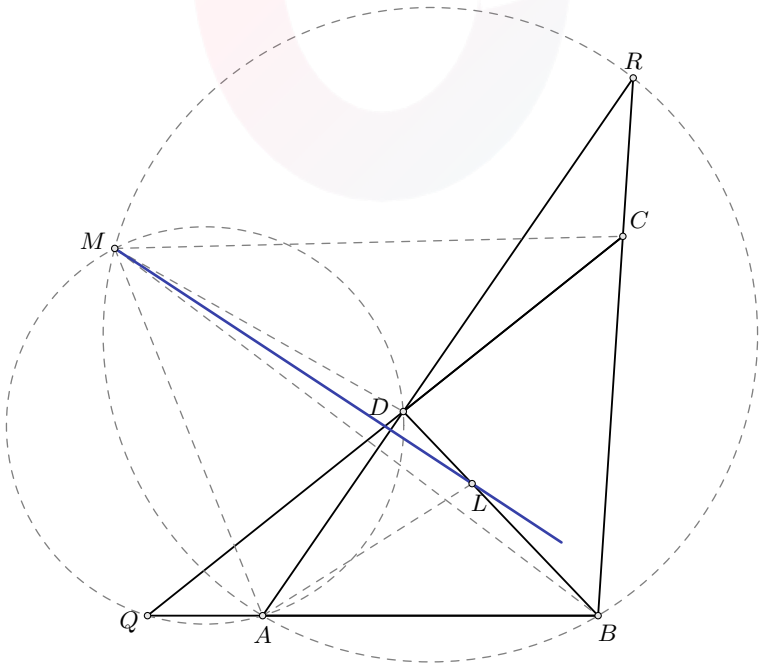
Step 3. Fixed shape. Let τ be the spiral similarity with centre S and $\tau(B) = D$. Maps with the same centre commute, so $\tau(A) = \tau(\sigma(B)) = \sigma(D) = E$. Thus τ sends the circle SBA to the circle SDE, which is the circle ADE, and sends the tangents at B and A to the tangents at D and E. Therefore $Z = \tau(Q)$, and triangle SDZ is directly similar to the fixed triangle SBQ.

By Lemma 1.4, $D \mapsto Z$ is a spiral similarity ρ with centre S and $\rho(B) = Q$. Also $\rho(A) = \rho(\sigma(B)) = \sigma(\rho(B)) = \sigma(Q) = P$. As D lies on line BA, the point $Z = \rho(D)$ lies on line QP. ■

Remark. The hypothesis is a statement about ratios on two segments, so a spiral similarity carries one segment to the other, here BA to AC. Once the circles ADE share the point S, the two extreme positions $D = B$ and $D = A$ name the line. By Theorem 3.2, S is the midpoint of the symmedian chord through A.

Problem 2

Let $ABCD$ be a convex quadrilateral with no two sides parallel and $AB \cdot CD = AD \cdot BC$. Lines AB and CD meet at Q, lines AD and BC meet at R, and the circles QAD and RAB meet again at M. Suppose that M lies on neither of the lines AC and BD. The A-symmedian of triangle ABD meets BD at L. Prove that line ML bisects both of the angles BMD and AMC.



Step 1. M is the Miquel point. By Theorem 2.1 the circles QAD and RAB both pass through the Miquel point M_0, which also lies on the circle RCD. The circle RCD meets line AD only at R and

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D, so $M_0 \neq A$. The circle RAB meets line AD only at R and A, so D is not on this circle and the circles QAD, RAB are different. Hence their second common point is M_0, and $M = M_0$.

Step 2. A ratio. By Theorem 2.1, M is the centre of a spiral similarity taking A, D to B, C and of another taking A, B to D, C. A spiral similarity multiplies all lengths by the same factor, so

$$\frac{MB}{MA} = \frac{BC}{AD}, \qquad \frac{MD}{MA} = \frac{DC}{AB}.$$

Dividing, and using $BC/DC = AB/AD$,

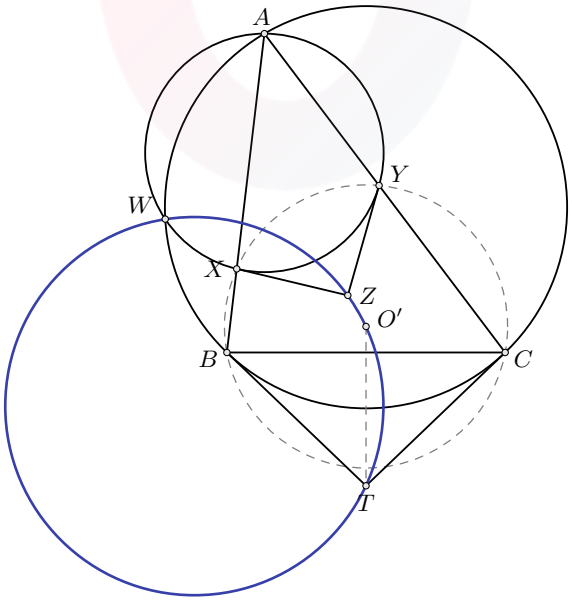
$$\frac{MB}{MD} = \frac{AB \cdot BC}{AD \cdot DC} = \frac{AB^2}{AD^2}.$$

Step 3. By Theorem 3.1(1) in triangle ABD, the point L lies on segment BD and $BL/LD = AB^2/AD^2 = MB/MD$. As M is not on line BD, the converse of the angle bisector theorem shows that ML is the internal bisector of angle BMD. By Theorem 2.3 the internal bisector of angle AMC is the same line. ■

Remark. For every quadrilateral the Miquel point sees B and D in the ratio $MB/MD = (AB \cdot BC)/(AD \cdot DC)$. The hypothesis turns this ratio into the square of a ratio of two sides, and such a square belongs to a symmedian. The same computation shows that ML passes through the foot on AC of the B-symmedian of triangle ABC.

Problem 3

Let ABC be a triangle with $AB \neq AC$, $\angle A \neq 90°$ and circumcircle Γ. A circle with centre O' passing through B and C meets the sides AB and AC again at X and Y. The circle AXY meets Γ again at W. The tangents to Γ at B and C meet at T, and the tangents to the circle AXY at X and Y meet at Z. Prove that O', T, Z, W lie on a circle.



Since B, C, Y, X are concyclic, XY is antiparallel to BC. So XY is parallel to the tangent to Γ at A, and this tangent is not parallel to BC, as $AB \neq AC$. Hence $XY \not\parallel BC$. The circles AXY and Γ do not touch at A, since a homothety at A taking one to the other would take XY to BC. So they meet at a second point $W \neq A$.

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Step 1. A spiral similarity with centre W. Let σ be the spiral similarity $X \mapsto B$, $Y \mapsto C$, which exists because $XY \nparallel BC$. Lines XB and YC meet at A, so by Theorem 1.2(2) the centre of σ lies on the circles AXY and ABC. The centre is not A. Otherwise σ would send X to the point B of ray AX and Y to the point C of ray AY, so σ would be a homothety and $XY \parallel BC$. Hence the centre is W.

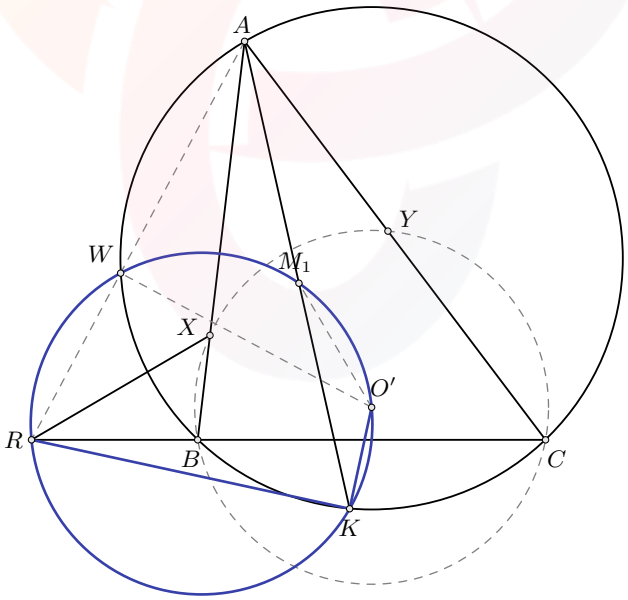
Step 2. Images. The map σ fixes W and sends X, Y to B, C, so σ sends the circle WXY to the circle WBC, that is, the circle AXY to Γ. Tangents go to tangents, so $\sigma(Z) = T$. Also σ sends the perpendicular bisector of XY to the perpendicular bisector of BC. The first line passes through Z, because $ZX = ZY$, and the second passes through T.

Step 3. Both perpendicular bisectors pass through O', and they are different lines because $XY \nparallel BC$. So line ZO' and its image line TO' meet at O', and σ takes the point Z of the first line to the point T of the second. By Theorem 1.2(1), the points W, O', Z, T are concyclic. ■

Remark. Two circles through A with a line through A on each side give a spiral similarity centred at their second common point, here W, the Miquel point of $BCYX$. The points Z and T are built in the same way from the two circles, so they correspond, and a line and its image always meet on a circle through the centre.

Problem 4

Let ABC be a triangle with $AB \neq AC$ and circumcircle Γ, and let the A-symmedian meet Γ again at K. A circle with centre O' passing through B and C meets the sides AB and AC again at X and Y. Lines XY and BC meet at R. Prove that $\angle O'KR = 90°$.



The segment XY is antiparallel to BC, so XY is parallel to the tangent to Γ at A, and this tangent is not parallel to BC because $AB \neq AC$. Hence R exists. Both X and Y lie on the same side of BC, so R is outside segment XY.

Step 1. The midpoint of XY. Let M_1 be the midpoint of XY. By Theorem 3.1(4), M_1 lies on line AK, between A and K. The inscribed angles give $\angle AKB = \angle ACB$, and the cyclic quadrilateral $BCYX$ gives $\angle AXY = \angle ACB$. So $\angle AXM_1 = \angle M_1KB$, an exterior angle of $XBKM_1$ equals the opposite interior angle, and X, B, K, M_1 are concyclic. The power of A gives

$$AM_1 \cdot AK = AX \cdot AB.$$

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Step 2. The Miquel point. The quadrilateral $BCYX$ is convex and cyclic with centre O', lines BX and CY meet at A, and lines BC and XY meet at R. Let W be the Miquel point. By Theorems 2.1 and 2.2, W lies on the circle RXB and on line AR, and $O'W \perp AR$. The power of A in the circle RXB gives

$$AW \cdot AR = AX \cdot AB.$$

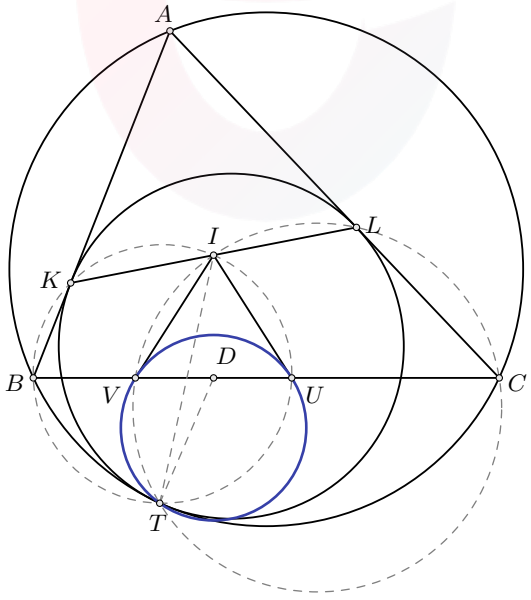
Step 3. From the two displays, $AM_1 \cdot AK = AW \cdot AR$. Lines AK and AR are different, since otherwise line XY would pass through A. By the converse of the power of a point, W, R, M_1, K are concyclic.

The point O' is not on Γ. The only points of Γ on the perpendicular bisector of BC are the two arc midpoints, and a circle through B and C centred at one of them does not meet both sides AB, AC again (for the arc BAC the power of A is negative, and for the other arc the second intersections are at distances AC and AB from A). So $O' \neq W$ and $\angle O'WR = 90°$. Hence the circle through W, R, M_1 is the circle on diameter $O'R$, because $O'M_1 \perp XY$ or $O' = M_1$. This circle contains K, and $K \neq O', R$. Therefore $\angle O'KR = 90°$. ■

Remark. A right angle at K over $O'R$ asks for the circle on diameter $O'R$. The centre O' projects to the midpoint of the chord XY and to the Miquel point on AR. The symmedian enters because K and the midpoint of every antiparallel lie on one line through A.

Problem 5

Let ABC be a triangle with incentre I, and let the incircle touch BC at D. The A-mixtilinear incircle touches AB, AC and the circumcircle at K, L and T. Points V and U lie on the side BC with $BV = BK$ and $CU = CL$. Prove that the lines IU and IV are tangent to the circle TUV, and that $\angle VTD = \angle ITU$.



Write A, B, C for the angles of the triangle. By Theorem 5.2(1), $KL \perp AI$ at I and $IK = IL$, so $\angle BKI = \angle CLI = 90° + \frac{A}{2}$.

Step 1. The points U and V. Reflection in BI swaps lines BA and BC, so this reflection takes K to V. Hence $IV = IK$ and $\angle BVI = 90° + \frac{A}{2}$. In the same way $IU = IL$ and $\angle CUI = 90° + \frac{A}{2}$. These

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angles are obtuse, so V lies between B and D, and U lies between D and C. Thus B, V, D, U, C lie in this order,

$$\angle IVU = \angle IUV = 90^\circ - \tfrac{A}{2},$$

and D is the midpoint of UV, because $IU = IV$ and $ID \perp BC$.

Step 2. Two circles. The points K and U lie on opposite sides of line BI, and $\angle BKI + \angle BUI = 180^\circ$. So U lies on the circle BKI, which contains T by Theorem 5.2(2). In the same way V lies on the circle $CLIT$.

Step 3. The angle UTV. The points I and T lie on opposite sides of BC. In the circle $BITU$ this gives $\angle BTU = 180^\circ - \angle BIU$, and in triangle BIU we have $\angle BIU = 180^\circ - \frac{B}{2} - \left(90^\circ - \frac{A}{2}\right)$. So $\angle BTU = 90^\circ + \frac{B}{2} - \frac{A}{2}$, and similarly $\angle VTC = 90^\circ + \frac{C}{2} - \frac{A}{2}$. Since $\angle BTC = 180^\circ - A$ and the rays TB, TV, TU, TC lie in this order,

$$\angle UTV = \angle BTU + \angle VTC - \angle BTC = 90^\circ - \tfrac{A}{2}.$$

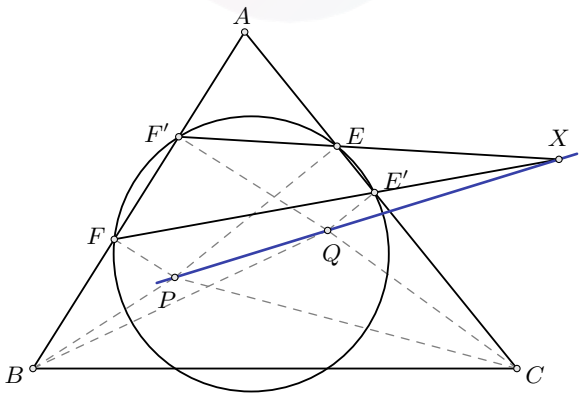
Step 4. Tangency. Now $\angle IVU = \angle VTU$ and $\angle IUV = \angle UTV$, with I and T on opposite sides of UV. By the converse of the tangent-chord theorem, IV and IU are tangent to the circle TUV.

Step 5. The tangents to the circle TUV at U and V meet at I, and $\angle UTV \neq 90^\circ$. By Theorem 3.1(3), TI is the T-symmedian of triangle TUV. The line TD is the median, so TD and TI are isogonal in angle UTV, and $\angle VTD = \angle ITU$. ■

Remark. The lengths $BV = BK$ and $CU = CL$ are reflections in BI and CI, which keep distances from I, so U, V, K, L lie on a circle with centre I. Tangents from I and a midpoint D then name a symmedian and a median. Since $\angle BTU - \angle BTI = \frac{B}{2}$, the last equality gives $\angle BTD = B = \angle ATC$.

Problem 6

Let ABC be an acute triangle. Points P and Q inside the triangle satisfy $\angle PBA = \angle QBC$, $\angle PCA = \angle QCB$ and $AP \neq AQ$. Let E and F be the feet of the perpendiculars from P to CA and AB, and let E' and F' be the feet of the perpendiculars from Q to CA and AB. Lines EF' and $E'F$ meet at X. Prove that X lies on line PQ.



Step 1. Isogonal conjugates. The lines BQ and CQ are isogonal to BP and CP, so $Q = P^*$ by Theorem 4.1. Hence AQ is isogonal to AP. Put $\alpha = \angle PAB = \angle QAC$ and $\beta = \angle PAC = \angle QAB$, both acute. Then $AF = AP\cos\alpha$, $AE = AP\cos\beta$, $AE' = AQ\cos\alpha$, $AF' = AQ\cos\beta$, and the four feet lie on the rays AB, AC, away from A. By Theorem 4.1 they lie on a circle ω whose centre N is the midpoint of PQ.

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If $EF' \parallel E'F$, then $AE/AE' = AF'/AF$, so $AE \cdot AF = AE' \cdot AF'$ and $AP = AQ$. So X exists. If $E = E'$, then $PQ \perp AC$ at E and $X = E$ lies on PQ. The same holds if $F = F'$. From now on the four feet are different.

Step 2. Three radical axes. Let c_P and c_Q be the circles on diameters AP and AQ. They pass through E, F and E', F'. The radical axis of c_P and ω is EF, and that of c_Q and ω is $E'F'$. The circles c_P and c_Q meet at A, and their centres are the midpoints of AP and AQ, so their radical axis is the line a through A perpendicular to PQ. The three radical axes are concurrent or parallel.

Step 3. EF and $E'F'$ meet at a point Y. Then Y lies on a, and $Y \neq A$. The points E, E', F, F' lie on ω, with $EE' \cap FF' = A$, $EF' \cap E'F = X$ and $EF \cap E'F' = Y$. By Brocard's theorem (L5-GE1, Theorem 4.4) the polar of X in ω is line AY, which is a. A polar is perpendicular to the line joining the pole to the centre, so $NX \perp a$. As $a \perp PQ$ and N lies on PQ, the point X lies on PQ.

Step 4. $EF \parallel E'F'$. Then a is parallel to both lines. The chords EF and $E'F'$ of ω have a common perpendicular bisector m through N. Reflection in m swaps E with F and E' with F', so m contains $X = EF' \cap FE'$ and $A = EE' \cap FF'$. Now m is perpendicular to a and passes through N, so m is line PQ. Again X lies on PQ. ■

Remark. Two angle conditions at B and C identify Q as the isogonal conjugate of P, which supplies a circle through the four feet with centre on PQ. A point defined by two chords of a circle is studied through its polar, and the right angles at the feet supply two more circles through A.